

Advanced topics in seasonal adjustment: Recap of introductory seasonal adjustment course

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■ Agenda

■ Motivation

■ RegARIMA pretreatment

- Regression equation
- Outlier effects
- Calendar effects
- ARIMA equation

■ The X11-Approach

- Basic algorithm
- Further issues

Why use seasonal adjustment?

Retail turnover in Germany

2015 = 100

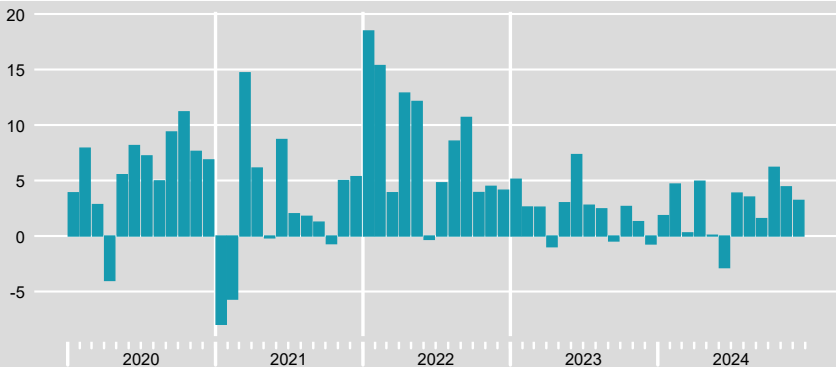


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A simple solution?

Retail turnover in Germany

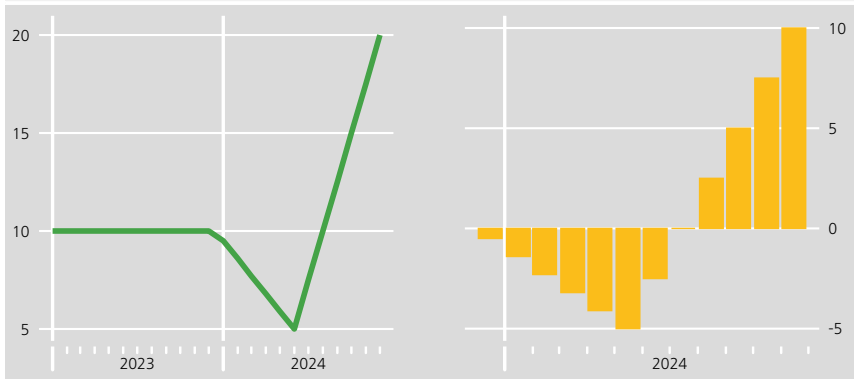
Year on year changes in %



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Some simulated data

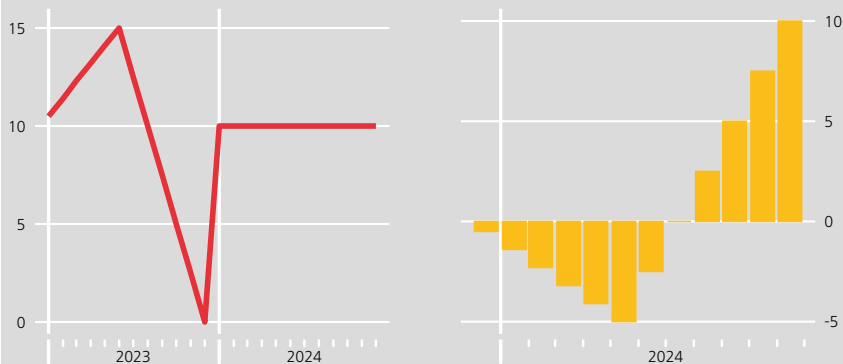
Time series A



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Another time series produces the same result

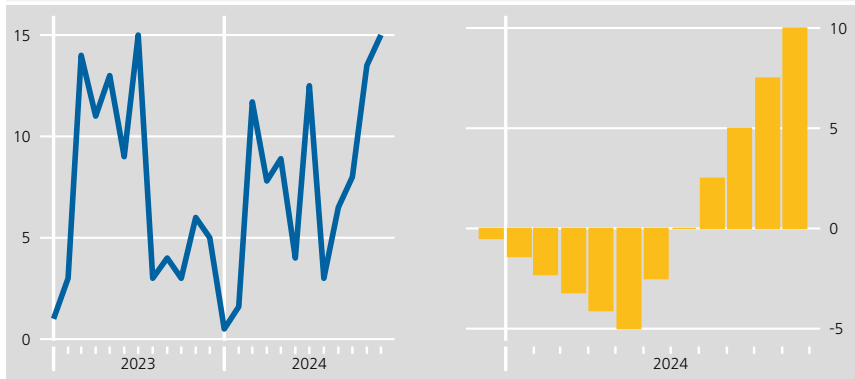
Time series B



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Still the same year-on-year-changes

Time series C

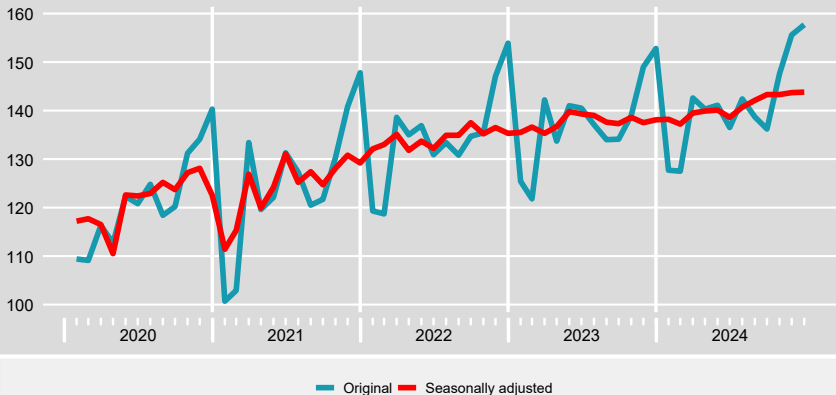


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Using proper seasonal adjustment

Retail turnover in Germany

2015 = 100



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■ A little quiz: What is seasonality?

1. Higher turnover in retail sales via Internet in November
2. Lower turnover in February compared to January & March
3. Higher energy consumption in leap year February
4. Higher turnover in retail sales of food during Ramadan
5. Higher orders received in August due to a biennial exhibition
6. Higher unemployment during winter

A little quiz: What is seasonality?

1. Higher turnover in retail sales via Internet in November ✓
2. Lower turnover in February compared to January & March ✓
3. Higher energy consumption in leap year February ✗
4. Higher turnover in retail sales of food during Ramadan ✗
5. Higher orders received in August due to a biennial exhibition ✗
6. Higher unemployment during winter ✓

■ The X-11 family

- X-11: Seasonal adjustment based on predefined linear filters
- X-11-ARIMA: Time series extension via ARIMA forecasts
- X-12-ARIMA: RegARIMA for outliers and calendar
- X-13-ARIMA-SEATS: Adaption to TRAMO and SEATS

Agenda

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RegARIMA pretreatment

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The X11-Approach

- Basic algorithm
- Further issues

Unobserved components model

Data

Calendar



recurring
movement

Seasonal



recurring
movement

Trend-cycle



long-term
growth path

Irregular



remaining
movements



Pre-Processing



X11-Core algorithm

Non-technical setup

$$\begin{array}{ccccccc} \text{Data} & = & \text{Outliers} & + & \text{Calendar variation} & + & \text{Disturbances} \\ & & & & \text{TD Holiday User} & & \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{A1} & & \text{A8} & & \text{A6 A7 A9} & & \text{B1} \end{array}$$

Technical setup

$$f(y_t) = \sum_k \beta_k \times k\text{-th regressor}_t + z_t$$

- y_t : Time series
- $f()$: Transformation function
- β_k : Outlier effects & calendar effects
- z_t ARIMA disturbances

Typical temporary outlier effects

- Atypical holiday constellations
- Extreme weather conditions
- Large-scale orders
- Major sport events
- Pandemics
- Strikes

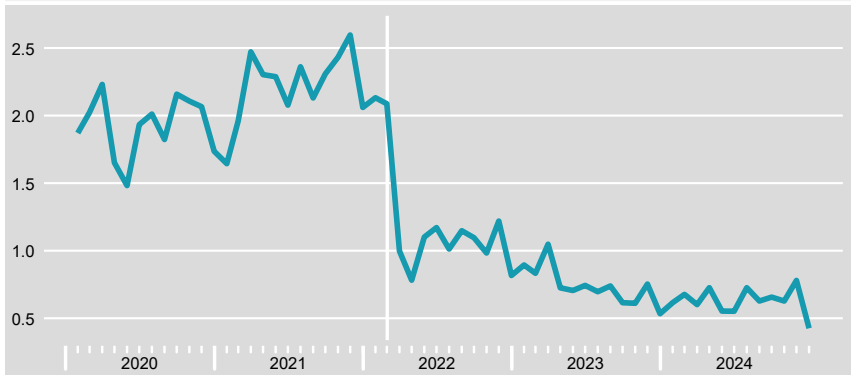
Typical permanent outlier effects

- Changes in data classification
- Legal changes
- Political changes

Example for outlier effects

German exports to Russia

In bn. €, vertical line marks beginning of Western sanctions in response to the Russia-Ukraine conflict

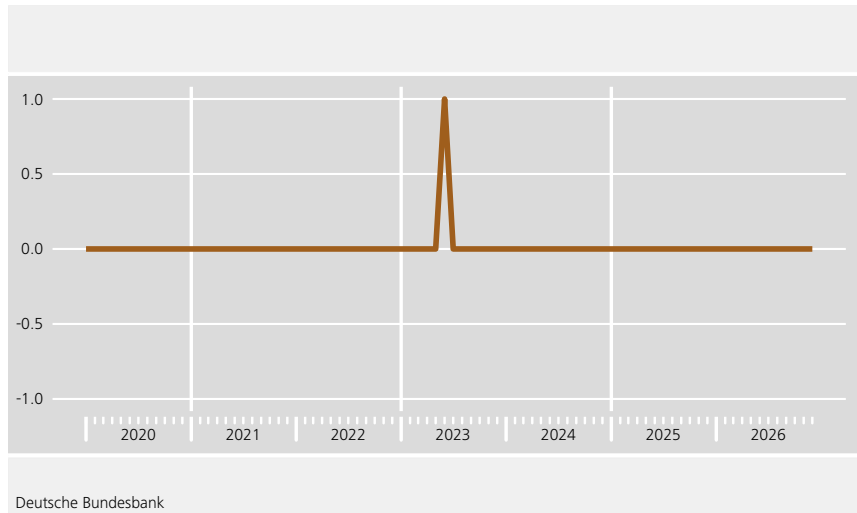


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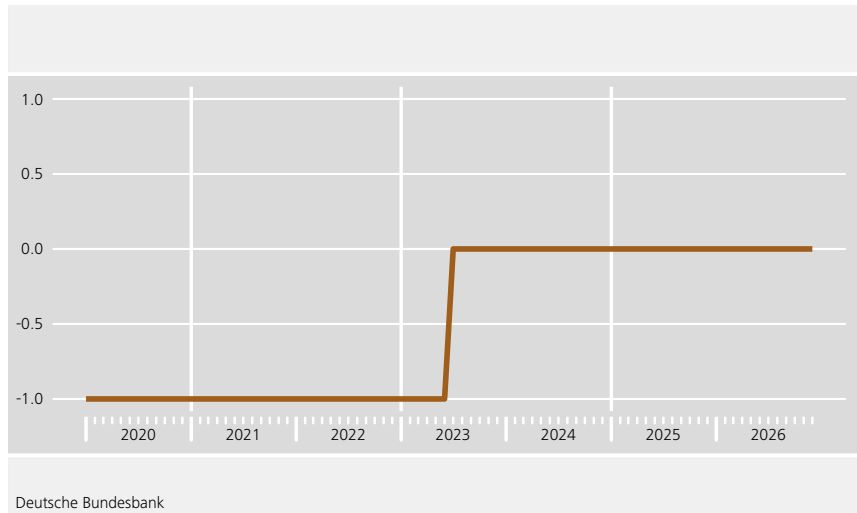
Outlier types

Effect	Outlier type	Component	Frequency of use
Temporary	Additive outlier	Irregular	Common
	Transitory change	Irregular	Occasional
Persistent	Level shift	Trend-cycle	Common
	Linear ramp	Trend-cycle	Rare
	Seasonal outlier	Seasonal	Rare

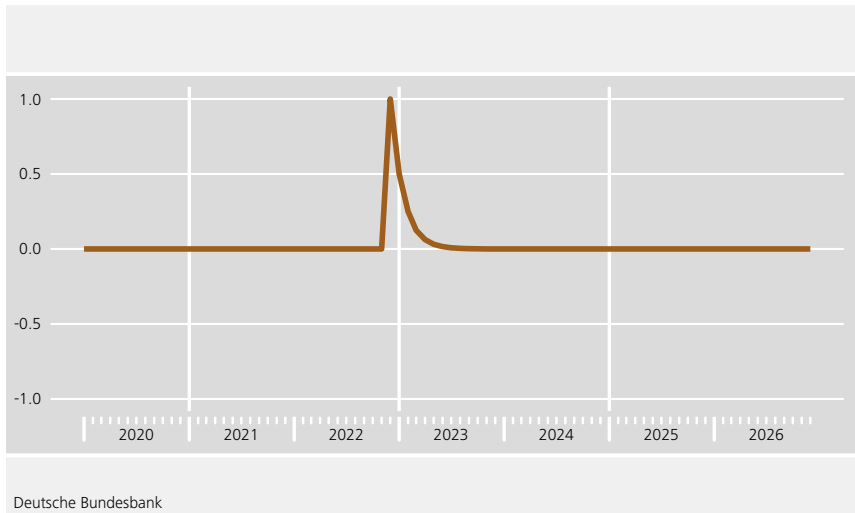
Additive outlier



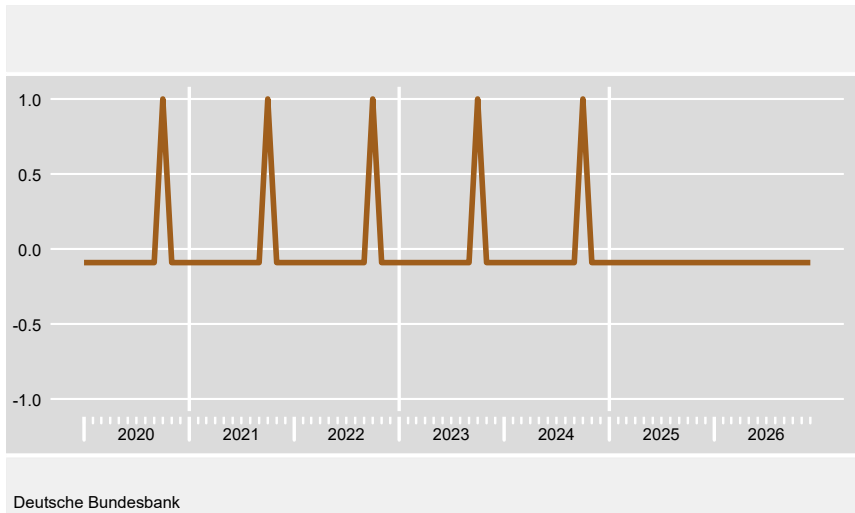
Level shift



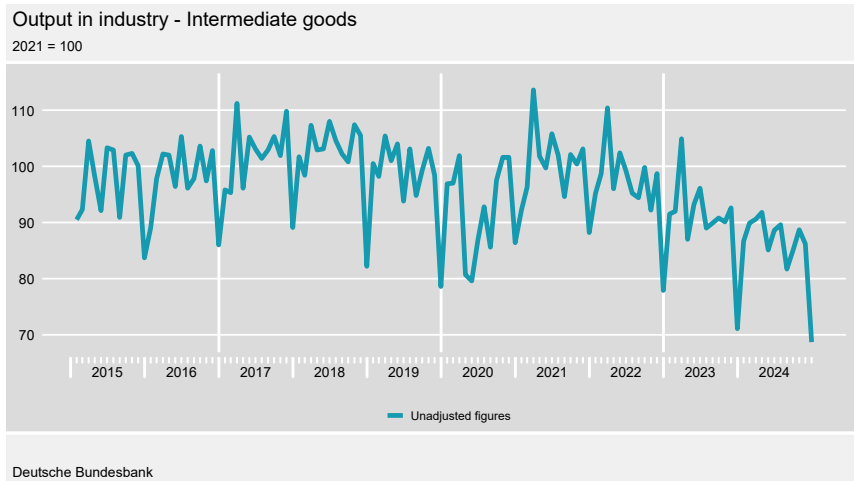
Transitory change



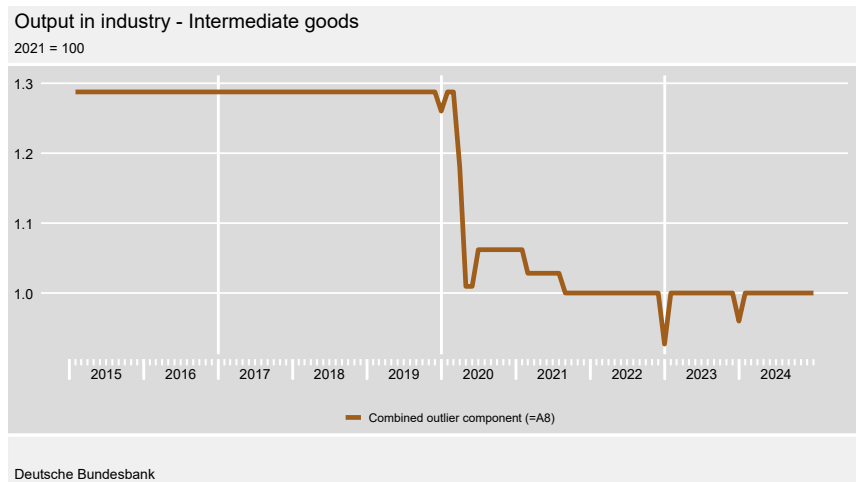
Seasonal outlier



Some raw data



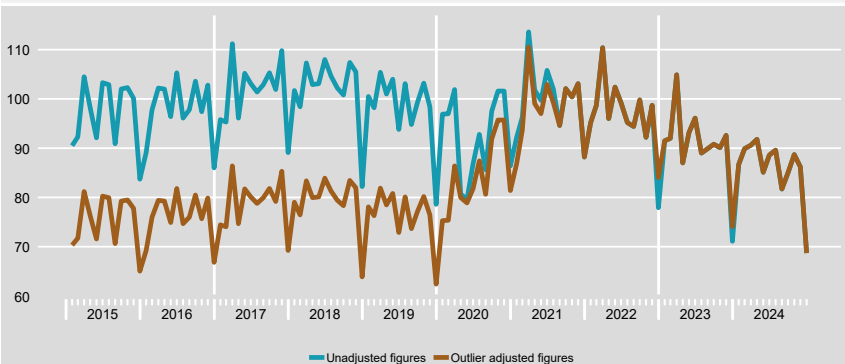
Estimated outlier component



Outlier adjustment

Output in industry - Intermediate goods

2021 = 100



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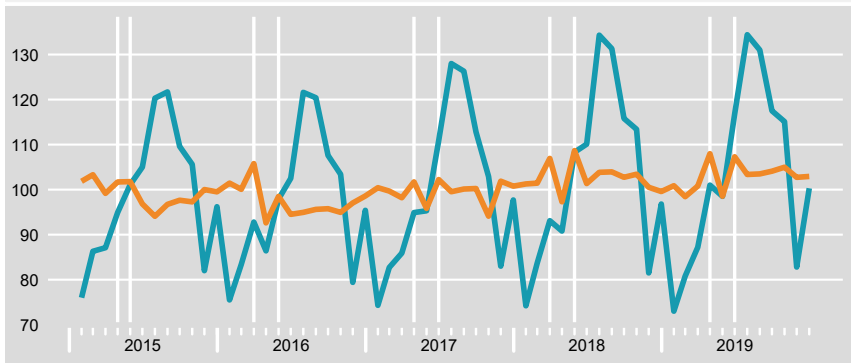
Typical calendar effects

- Number of working days in a month
- Number of trading days in a month
- Holidays, Bridge days, ...

Example for calendar effects

CPI package holidays

2015 = 100, vertical lines mark Easter and Pentecost holiday



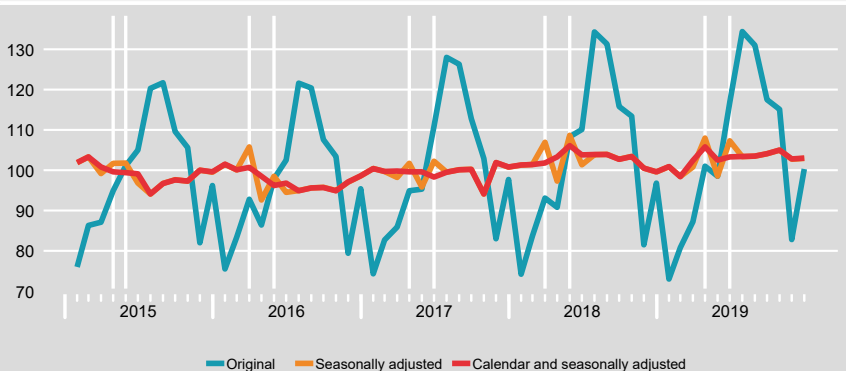
— Original — Seasonally adjusted

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Example for calendar effects

CPI package holidays

2020 = 100, vertical lines mark Easter and Pentecost holiday

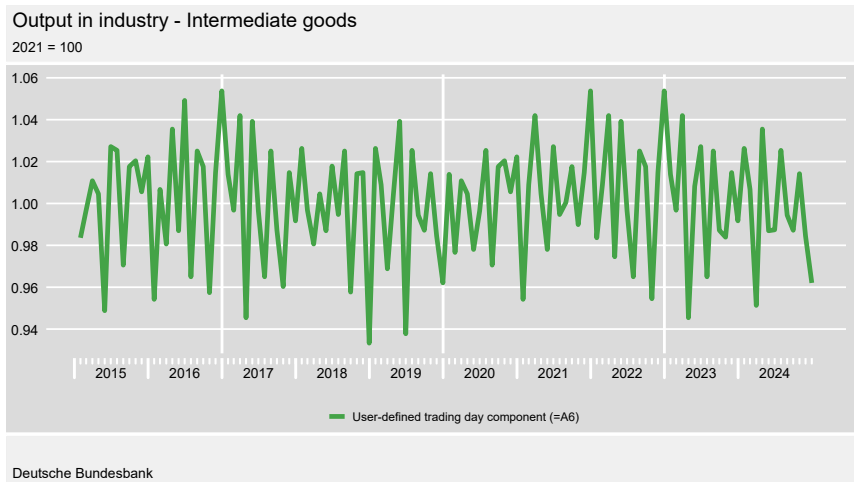


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I Default options for calendar regressors in JD+

- Number of working days in a month
- Number of trading days in a month
- Leap year effect
- Easter effect

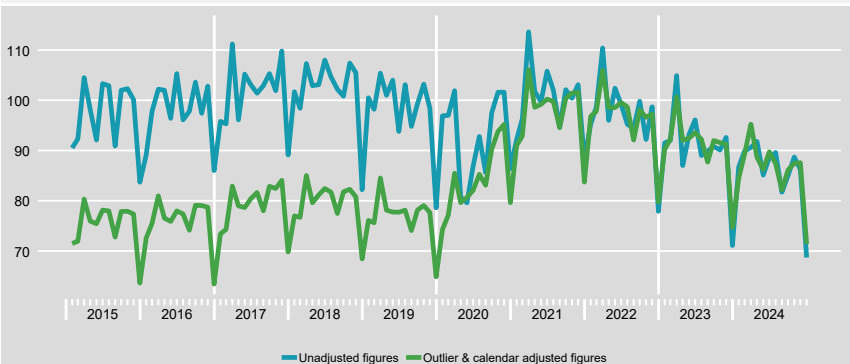
Custom trading day regressors



Calendar adjustment

Output in industry - Intermediate goods

2021 = 100



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Remember: Regression equation

Data = Outliers + Calendar variation + Disturbances

TD Holiday User



A1



A8



A6 A7 A9



B1

Non-technical setup

$$\begin{array}{ccccccccc} \text{AR} & \text{SAR} & \text{D} & \text{SD} & \text{B1} & = & \text{MA} & \text{SMA} & \text{WN} \\ \downarrow & \downarrow & \downarrow & \downarrow & & & \downarrow & \downarrow & \\ p & P & d & D & & & q & Q & \end{array}$$

Technical setup

AR	SAR	D	SD	B1 =	MA	SMA	WN
↓	↓	↓	↓	↓	↓	↓	↓
$\phi_p(B)$	$\Phi(B^{12})$	$(1 - B)^d$	$(1 - B^{12})^D$	$z_t =$	$\theta_q(B)$	$\Theta_Q(B^{12})$	ε_t

– B : Backshift operator: $B^k z_t = z_{t-k}$

– $f(pdq)(PDQ)$

The most popular ARIMA model

$$(1 - B)(1 - B^{12})z_t = (1 - \theta_1 B)(1 - \Theta_1 B^{12})\varepsilon_t$$

- $-1 \leq \theta_1, \Theta_1 \leq 1$
- $(pdq)(PDQ) = (011)(011)$ ARIMA

Summary of pre-processing

- Potential log transformation, dependant on log-level test
- Temporary removal of outliers
- Permanent removal of calendar variation
- Seasonal ARIMA assumption for disturbances

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Pre-Processing



X11-Core algorithm

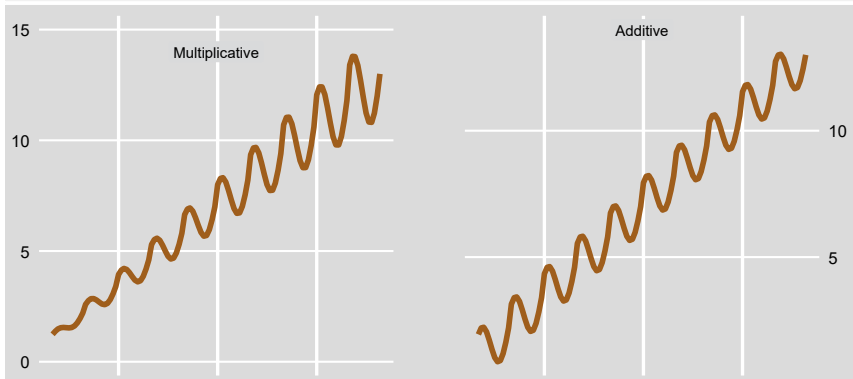
Technical Setup

$$\begin{array}{ccccccc} \text{Pre-treated data} & & \text{Seasonal} & & \text{Trend-cycle} & & \text{Irregular} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ z_t & = & s_t & \circ & t_t & \circ & i_t \end{array}$$

Two types of decomposition

Additive vs. multiplicative decomposition

Trend-cyclical and seasonal component



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Multiplicative model

$$\begin{array}{ccccccc} \text{Data} & & \text{Seasonal} & & \text{Trend-cycle} & & \text{Irregular} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ z_t & = & s_t & \times & t_t & \times & i_t \end{array}$$

$$\text{adjusted series} = z_t / s_t$$

Additive model

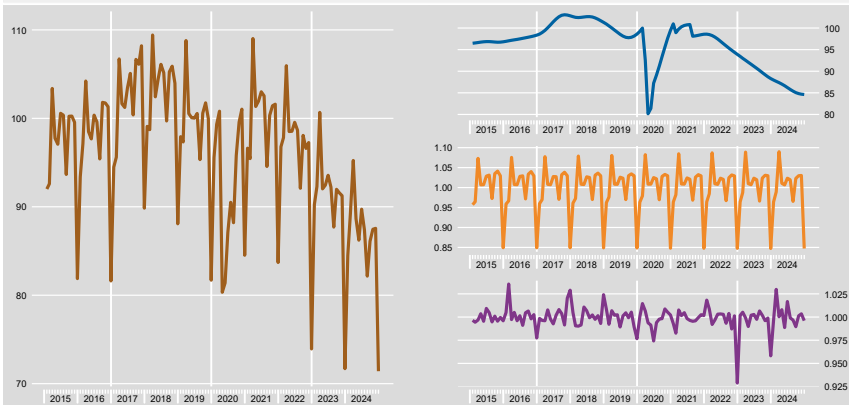
$$\begin{array}{ccccccc} \text{Data} & & \text{Seasonal} & & \text{Trend-cycle} & & \text{Irregular} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ z_t & = & s_t & + & t_t & + & i_t \end{array}$$

$$\text{adjusted series} = z_t - s_t$$

The components in action

Output in industry - Intermediate goods

Volume, 2021 = 100, left: calendar-adjusted, right: trend-cycle, seasonal component, irregular component



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Four repeated steps

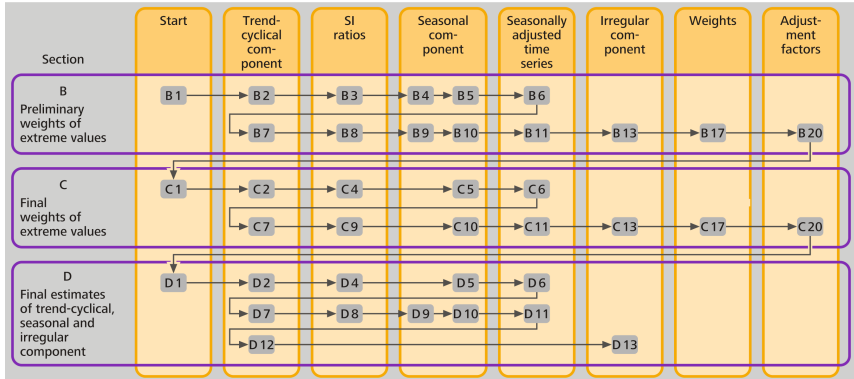
Data		Seasonal		Trend-cycle		Irregular
↓		↓		↓		↓
z_t	=	s_t	×	t_t	×	i_t

1. Estimate trend $\Rightarrow \hat{t}_t$
2. Remove trend from series $\Rightarrow z_t / \hat{t}_t = \widehat{s_t \cdot i_t}$
3. Estimate seasonal $\Rightarrow \hat{s}_t$
4. Remove seasonal from series $\Rightarrow z_t / \hat{s}_t = \widehat{t_t \cdot i_t}$

The complete X-11 algorithm

Basic principle of the X-11 seasonal adjustment algorithm in JDemetra+

Workflow diagram



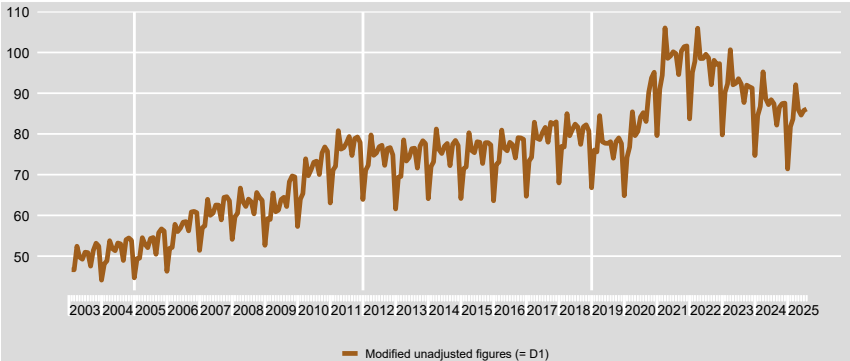
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S3PR0037C_Chart

Table D1

Output in industry - Intermediate goods

2021 = 100



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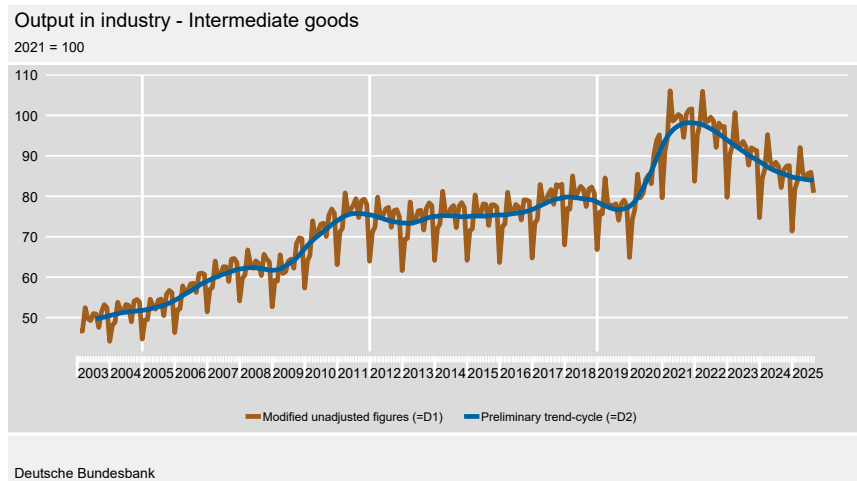
■ Step 1: Estimate (preliminary) trend

$$\hat{t}_t = \frac{1}{24}z_{t-6} + \frac{1}{12}z_{t-5} + \cdots + \frac{1}{12}z_t + \cdots + \frac{1}{12}z_{t+5} + \frac{1}{24}z_{t+6}$$

Moving average filters graphically

	1	2	3	4	5	6	7	8	9	10	11	12
year $n - 2$												
year $n - 1$									x	x	x	x
year n	x	x	X	x	x	x	x	x	x			
year $n + 1$												
year $n + 2$												

Table D2



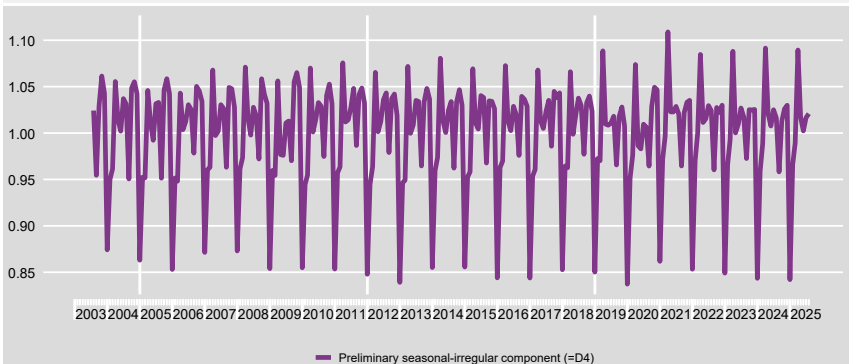
■ Step 2: Remove (preliminary) trend estimate

$$(\hat{s}i)_t = \frac{z_t}{\hat{t}_t}$$

Table D4

Output in industry - Intermediate goods

2021 = 100



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Step 3: Estimate (preliminary) seasonal component

$$\hat{s}_t = \frac{1}{9}(\hat{s}i)_{t-24} + \frac{2}{9}(\hat{s}i)_{t-12} + \frac{3}{9}(\hat{s}i)_t + \frac{2}{9}(\hat{s}i)_{t+12} + \frac{1}{9}(\hat{s}i)_{t+24}$$

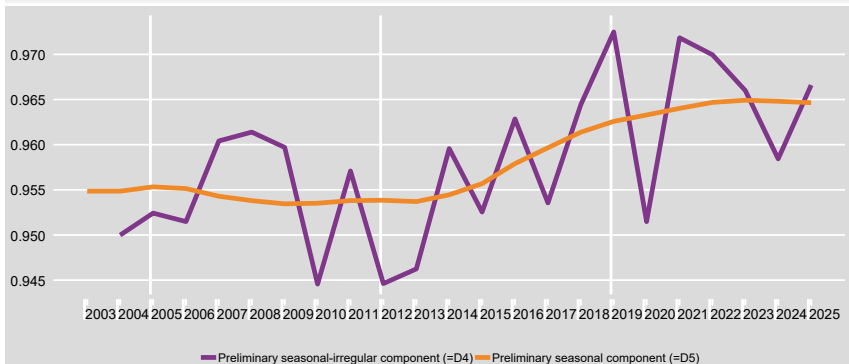
Seasonal moving average filters graphically

	1	2	3	4	5	6	7	8	9	10	11	12
year $n - 2$			x									
year $n - 1$			x									
year n			X									
year $n + 1$			x									
year $n + 2$			x									

SI-Ratios and preliminary seasonal components

Output in industry - Intermediate goods

2021 = 100, only January values

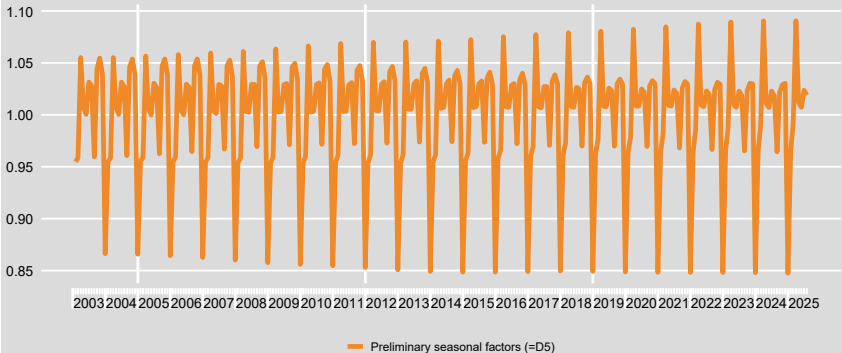


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Table D5

Output in industry - Intermediate goods

2021 = 100



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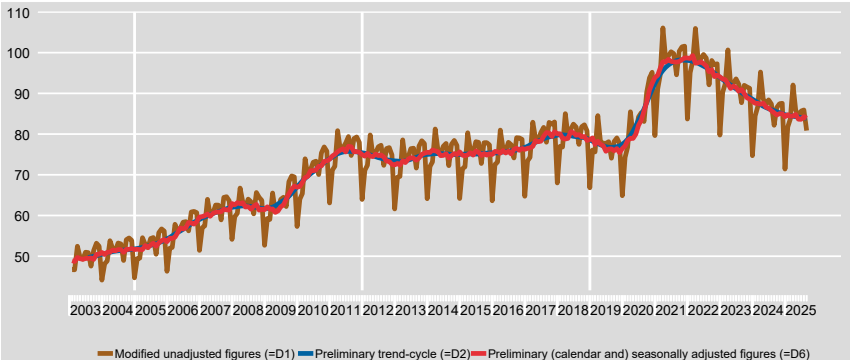
■ Step 4: Remove (preliminary) seasonal estimate

$$\hat{z}_t^{(sa)} = \frac{z_t}{\hat{s}_t}$$

Table D6

Output in industry - Intermediate goods

2021 = 100

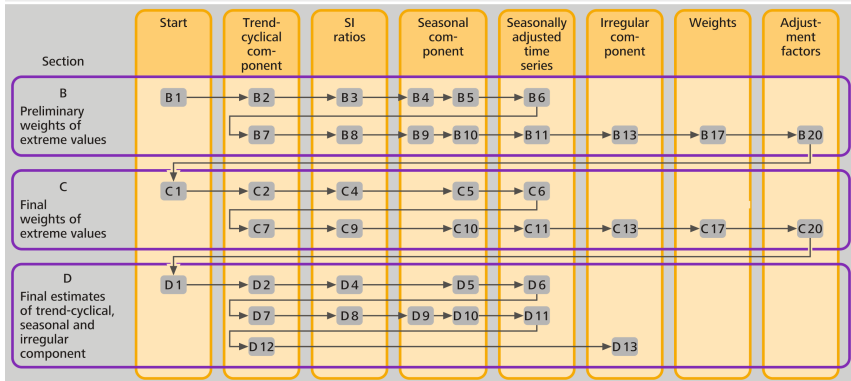


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Recap: The X-11 algorithm

Basic principle of the X-11 seasonal adjustment algorithm in JDemetra+

Workflow diagram



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S3PR0037C_Chart

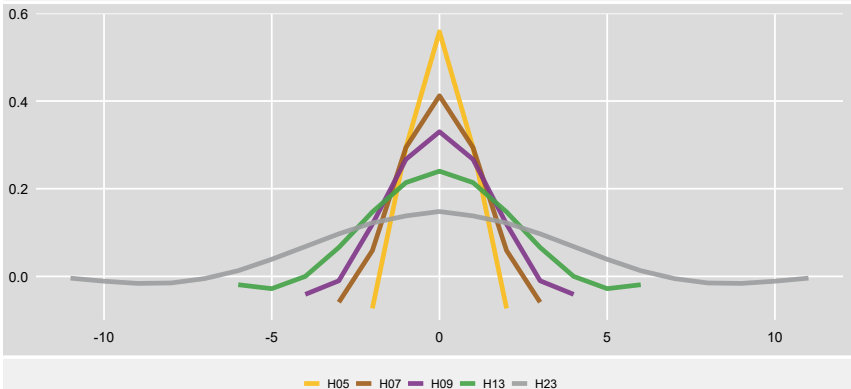
Standard centred MA-12 filter

$$\hat{t}_t = \frac{1}{24}z_{t-6} + \frac{1}{12}z_{t-5} + \cdots + \frac{1}{12}z_t + \cdots + \frac{1}{12}z_{t+5} + \frac{1}{24}z_{t+6}$$

Henderson filters

Weights of Henderson filters

Filters: 5-term, 7-term, 9-term, 13-term, 23-term

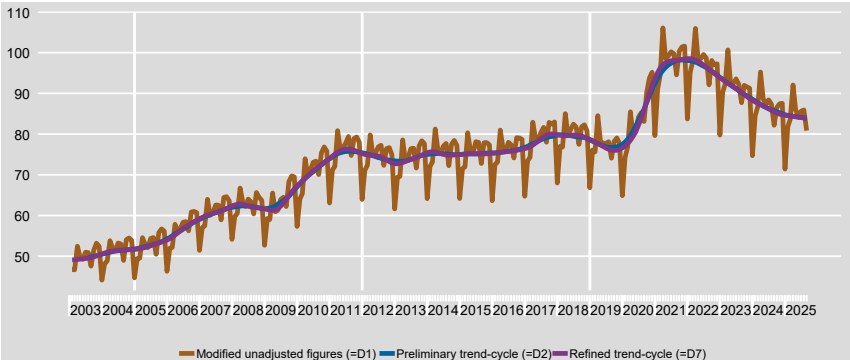


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Table D7

Output in industry - Intermediate goods

2021 = 100



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■ SI-Ratio with extreme value replacement

$$(\hat{s}i)_t = \frac{z_t}{\hat{t}_t}$$

- $\mathbb{E}(i_t) = 0$ when additive
- $\mathbb{E}(i_t) = 1$ when multiplicative
- Lower and upper limits: L and U
- Normal if $\hat{i}_t \in [\mu_i - L \cdot \sigma_t, \mu_i + L \cdot \sigma_t]$ with μ = mean and σ = standard deviation

Extreme SI-ratios

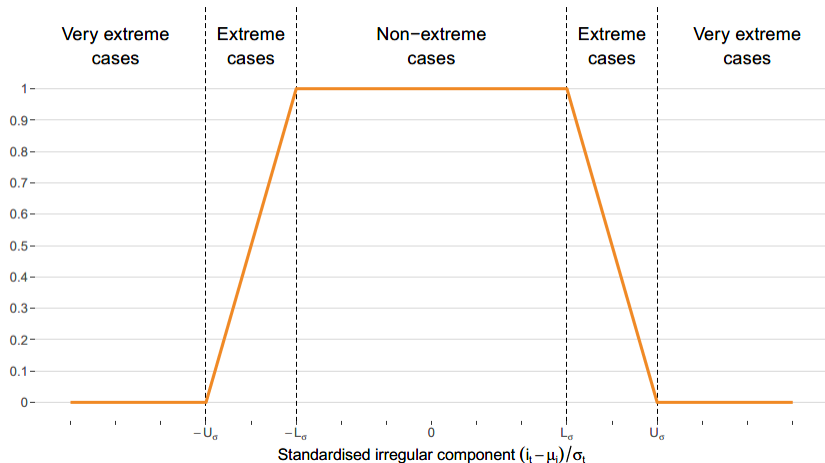
– Too small:

- Extreme: $\hat{i}_t \in (\mu_i - U \cdot \sigma_t, \mu_i - L \cdot \sigma_t)$
- Very Extreme: $\hat{i}_t \in (-\infty, \mu_i - U \cdot \sigma_t]$

– Too large:

- Extreme: $\hat{i}_t \in (\mu_i + L \cdot \sigma_t, \mu_i + U \cdot \sigma_t)$
- Very Extreme: $\hat{i}_t \in [\mu_i + U \cdot \sigma_t, +\infty)$

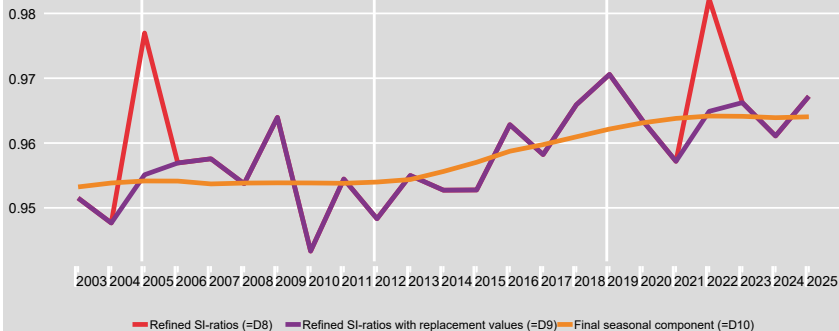
Weighting function for extreme values



Final SI-Ratios and final seasonal components

Output in industry - Intermediate goods

2021 = 100, only January values

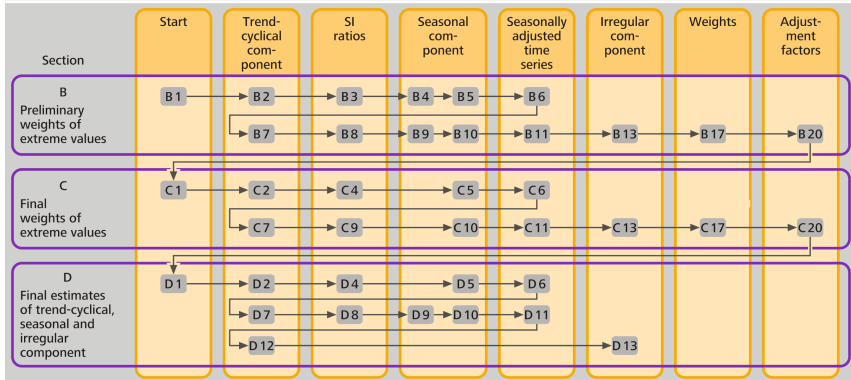


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We completed the algorithm

Basic principle of the X-11 seasonal adjustment algorithm in JDemetra+

Workflow diagram



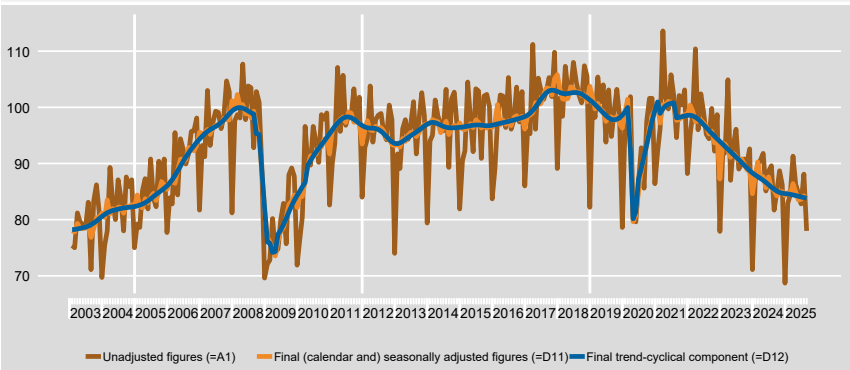
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S3PR0037C_Chart

Tables D11 & D12

Output in industry - Intermediate goods

2021 = 100, only January values



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